

6(a)(i)	any two from <table><tr><th>structural gene</th><th>name of gene product</th></tr><tr><td><i>lacZ</i></td><td>β-galactosidase / lactase ;</td></tr><tr><td><i>lacY</i></td><td>(lactose) permease ; A (β-galactoside) permease</td></tr><tr><td><i>lacA</i></td><td>transacetylase ;</td></tr></table>	structural gene	name of gene product	<i>lacZ</i>	β -galactosidase / lactase ;	<i>lacY</i>	(lactose) permease ; A (β -galactoside) permease	<i>lacA</i>	transacetylase ;	2
structural gene	name of gene product									
<i>lacZ</i>	β -galactosidase / lactase ;									
<i>lacY</i>	(lactose) permease ; A (β -galactoside) permease									
<i>lacA</i>	transacetylase ;									
6(a)(ii)	gene products / enzyme / protein, needed all the time ;	1								
6(a)(iii)	any two from 1 (repressor protein / gene product), binds to the operator ; 2 blocks promoter ; 3 RNA polymerase unable to bind to promoter ; 4 no, transcription / expression / activation / mRNA synthesis, of (named) structural genes ;	2								

Question	Answer	Marks
6(b)(i)	1 only produced when, substrate / inducer / lactose, is present ; 2 (substrate / inducer / lactose) causes gene expression / gene activation / transcription / mRNA synthesis ;	2
6(b)(ii)	no waste of, amino acids / ATP / nucleotides / energy ;	1

Question	Answer	Marks
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